EC234AI

USN | 1 | R | V | 2 | 2 | E | C | 1 | 7 | 8 |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Examinations Apr/May-2024

Common to ECE/EEE/EI/ETE

ANALYSIS AND DESIGN OF DIGITAL CIRCUITS WITH HDL

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

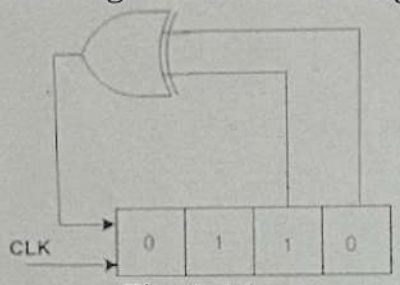
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	If reg [0:31] a_bus,, then a_bus[20-:8] is equal to _____.	01	2	2
	1.2	_____ type of case statement will consider both x and z as don't care.	01	1	3
	1.3	If w1 = 4'b101x and y1 = 4'b010x then w1==y1 equals _____.	01	2	4
	1.4	For the given code If a= 1'b1, en= 1'b1, then y1 equals module test (input a, en, output y1); tri0 y1; bufif0 (y1, a, en); endmodule	01	3	1
	1.5	In a J-K flip flop if, J=Q' and K=1. Assuming that flip flop is initially cleared and then clocked for 6 pulses, the sequence at the Q output will be _____.	02	2	2
	1.6	In 8:3 priority encoder circuit, if the input lines are having the priority in the order D7 to D0, if the input line status is 01XXXXXX, the output will be _____(Assume active low outputs)	01	3	3
	1.7	The sequential circuit consists of 3 FFs A, B, C and 2 inputs x and y. The flip-flop C is a JK flip flop and defined by state equation C(t+1)= Cx+AB. The input of flip-flop C are J= _____ and K= _____.	02	3	3
	1.8	The initial state of 8-bit shift register is 11110000. The data 10110111 needs to be serially loaded (LSB first) into the register. After four clock pulses, the register contains _____.	01	1	1

	1.9	What is the duty cycle of clock generated by following block of code
		initial
		begin
		clk=1'b0;
		forever
		begin
		#4 clock=1'b1;
		#6 clock=1'b0;
		end
			01	2	3

1.10	For the following Verilog code segment, if the initial value of "din" is 32'h3456ABCD, the value of "data" in decimal will be
	(Note that "data" is a 4-bit variable)
	wire [31:0] din;
	wire [3:0] data;
	wire [15:0] d1;
	wire [31:16] d2;
	assign d1 = din[31:16];
	assign d2 = din[15:0];
	assign data = d1[7:4]|d2[23:20]|d2[27:24];
		01	2	3

1.11	In a BCD adder, what happens when the sum of two BCD digits is greater than 9?	01	1	1
1.12	The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output shift register withan initial state 01110. After three clock pulses, the register contains _____.	01	1	2
1.13	A sequence detector is set to detect the sequence 1010. How many times the output will go high if a stream of 1010101110010101011 is sent as an input and overlapping sequence is countable?	01	2	4
1.14	In a 4-bit right shift register given in Fig. 1.14, how many clock pulses are required to change the content of register all 1's.  Fig. 1.14	02	3	4
1.15	A 4-bit serial in parallel out shift register is initially set to 1111. The data 1010 is applied to the input. After 3 clock pulses the output will be _____.	01	2	3
1.16	For the following Verilog module the output 'f' equal to _____. module Q1 (f, a, b, c); input a, b, c; output f, wire t1, t2; nand G1 (t1, a, b); nor G2 (t2, b, c); or G3 (f, t1, t2); endmodule	02	3	4

PART-B

2	a	Differentiate between the following i) bufif1, notif0 3 ii) Wire, Wor and reg 3	06	1	1
	b	Design half-adder using "nand" gates only. Write Verilog structural code for same. Design 1-bit full adder using half adders and write Verilog code. Design a 4-bit binary adder/subtractor using 1-bit full adders and write the Verilog structural code for same. 3+3+4	10	3	3
3	a	Implement the Boolean function F(A,B,C,D)= $\sum m(1,2,3,4,11,12,13,14,15)$ using 8:1 multiplexer. Show the design procedure.	06	3	1
	b	Using for loop write Verilog code for n-bit ripple carry adder.	10	2	2

	OR			
4 a	Design 1-bit magnitude comparator with output A>B, A<B, A=B. Write a Verilog code using dataflow model.	08	3	3
b	Design 2-bit magnitude comparator using 1 comparator and required basic gates; Write a Verilog code for same using structural modeling.	08	3	3
5 a	Design the working module of Synchronous Binary 8 bit Up-Down counter, which performs the operation at positive edge of clock (CLK) and develop a behavioral description in Verilog using if statement considering input as DATA of eight bit and counter output as DOUT of eight bit. i) Set the counter to an initial state through input DATA when SET is high. ii) Clearing the counter to zero (0) state when CLR is high. iii) Increment the counter when DIR is high and both SET and CLR are low. iv) Decrement the counter when all DIR, SET and CLR are low.	06	2	4
b	Design a synchronous counter using T-FFs such that it counts the sequence 1-3-5-7-11-13-15-1-3-5-7..... and so on. Avoid lock out condition in the design.	10	4	4
	OR			
6 a	Design a sequential circuit using D-FFs satisfying the following state conditions with the help of state table and state diagram: Given X and Y are the input and output respectively. A(t+1)= A(t)'. X+B(t). X' B(t+1)= A(t). X' Y(t)= (A(t)+B(t))'.X	10	4	3
b	Write a Verilog code for JK flipflop using case statement and write a testbench to verify the same.	06	2	2
7 a	Draw the state diagram of a circuit that detects the following sequence: 1011010. The detector must assert output 'z=1' when the sequence is detected. (With Overlapp & Mealy machine). Develop a verilog code for the designed FSM.	08	3	3
b	Construct a Johnson counter using D-FF. Write a Verilog code and explain with timing diagram.	08	2	2
	OR			
8 a	Write a Behavioral code for a 16-bit universal shift register. Specifications for a 16-bit shift register are as follows: I/O description: clk= Positive-Edge Clock Synchronization ; n_rst= Active low synchronous reset input; SDL : Serial data left; SDR: Serial data right, S[2:0]: Define operation of Shift register as defined below; S_Load: Synchronous Parallel Load (active High); D[15:0]: Parallel data output Operations: 000= No Action; 001= Register is shifted right and serial data SDR enters; 010= Register is shifted left and serial data SDL enters; 011= Logical shift right; 100= Logical shift left; 101= Arithmetic shift right; 110= Arithmetic shift left; 111= Parallel load.	08	3	3

	b	Consider the case of a circuit to detect a pair of 1's or 0's in the single bit input. That is, input will be a series of one's and zero's. If two one's or two zero's comes one after another, output should go high. Otherwise output should be low. Develop a Verilog Code for the circuit using Mealy FSM.	08	3	3
9	a	What is scratch pad memory? Explain the organization of processor unit using scratch pad memory with the help of a neat diagram.	06	2	4
	b	Design an arithmetic and logic unit with three selection variables $S_2S_1S_0$ that generate the following operations. Draw the logic diagram of one typical stage.	10	1	3

Arithmetic Operations					Logical Operations			
S_2	S_1	S_0	$C_{in}=0$	$C_{in}=1$	S_2	S_1	S_0	$C_{in}=X$
0	0	0	F=A	F=A+1	1	0	0	OR
0	0	1	F=A+B	F=A+B+1	1	0	1	XOR
0	1	0	F=A-B-1	F=A-B	1	1	0	AND
0	1	1	F=A-1	F=A	1	1	1	Complement

OR

10	a	With a neat diagram explain i) Bus organization with respect to processor ii) Accumulator register	08	2	4
	b	With a neat block diagram, explain operation of a status register.	08	1	3